Autocorrelation

EC 421, Set 8

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Prologue

Schedule

Last Time

Introduction to time series

Today

Autocorrelation

R showcase

Functions

Writing your own functions.

Writing functions

Functions are everywhere

Everything you do in R involves some sort of function, *e.g.*,

- `mean()`
- `lm()`
- `summary()`
- `read_csv()`
- `ggplot()`
- `+`

The basic idea in R is doing things to objects with functions.

Writing functions

Functions can help

We write functions to make life easier. Instead of copying and pasting the same line of code a million times, you can write one function.

In R, you use the `function()` function to write functions.[1]

```
# Our first function
the_name <- function(arg1, arg2) {
  # Insert code that involves arg1 and arg2 (this is where the magic happens)
}
```

- `the_name`: The name we are giving to our new function.
- `arg1`: The first argument of our function.
- `arg2`: The second argument of our function.

1. Meta.

Writing functions

Our first real function

Let's write a function that multiplies two numbers. (It needs two arguments.)

```
# Create our function
the_product <- function(x, y) {
  x * y
}
```

Writing functions

Functions can do anything

... that you tell them.

If you are going to repeat a task (*e.g.*, a simulation), then you have a good situation for writing your own function.

R offers many functions (via its many packages), but you will sometimes find a scenario for which no one has written a function.

Now you know how to write your own.

```
# An ad lib function
ad_lib <- function(noun1, verb1, noun2) {
  paste("The next", noun1, "of our lecture", verb1, noun2)
}
```

Writing functions

```r
ad_lib(noun1 = "part", verb1 = "reviews", noun2 = "time series.")
```

```
[1] "The next part of our lecture reviews time series."
```

Time series

Review

Time series

Review

Changes to our model/framework.

- Our model now has t subscripts for **time periods**.
- **Dynamic models** allow **lags** of explanatory and/or outcome variables.
- We changed our **exogeneity** assumption to **contemporaneous exogeneity**, *i.e.*, $\boldsymbol{E}\left[u_t | X_t\right] = 0$
- Including **lags of outcome variables** can lead to **biased coefficient estimates** from OLS.
- **Lagged explanatory variables** make **OLS inefficient**.

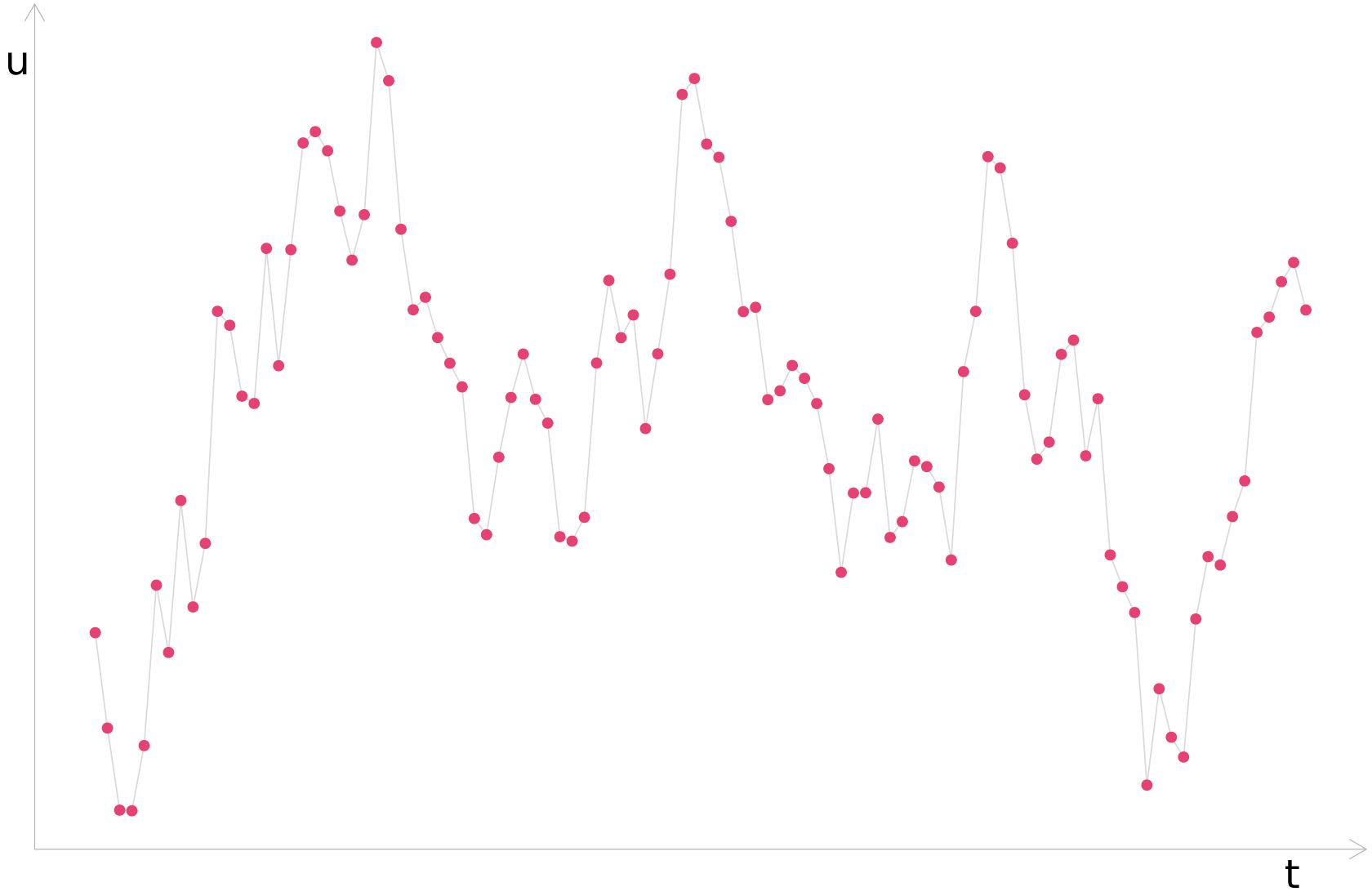
Autocorrelation

Autocorrelation

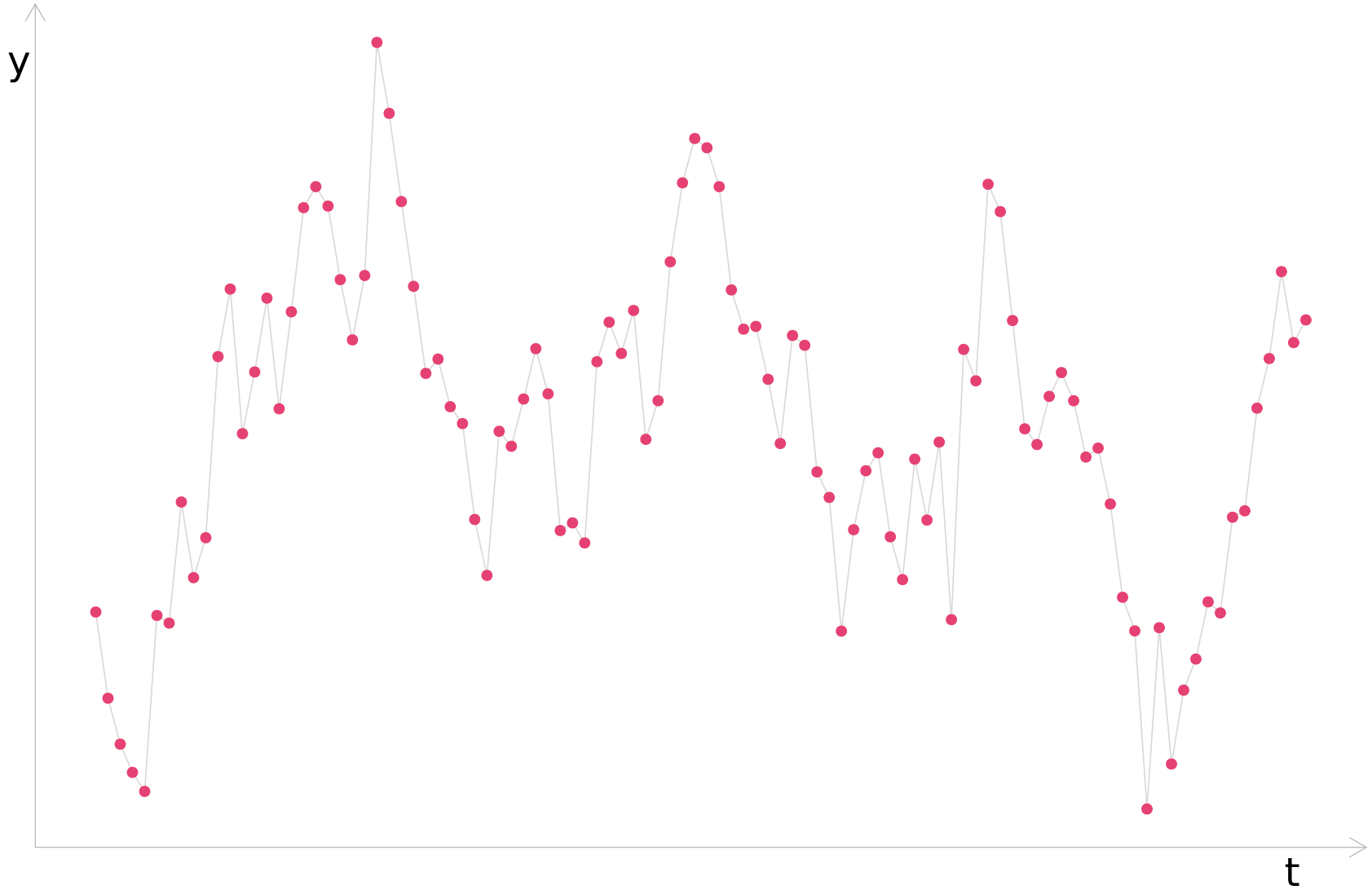
What is it?

Autocorrelation occurs when our disturbances are correlated over time, *i.e.*, $\text{Cov}\left(u_t,\, u_s\right) \neq 0$ for $t \neq s$.

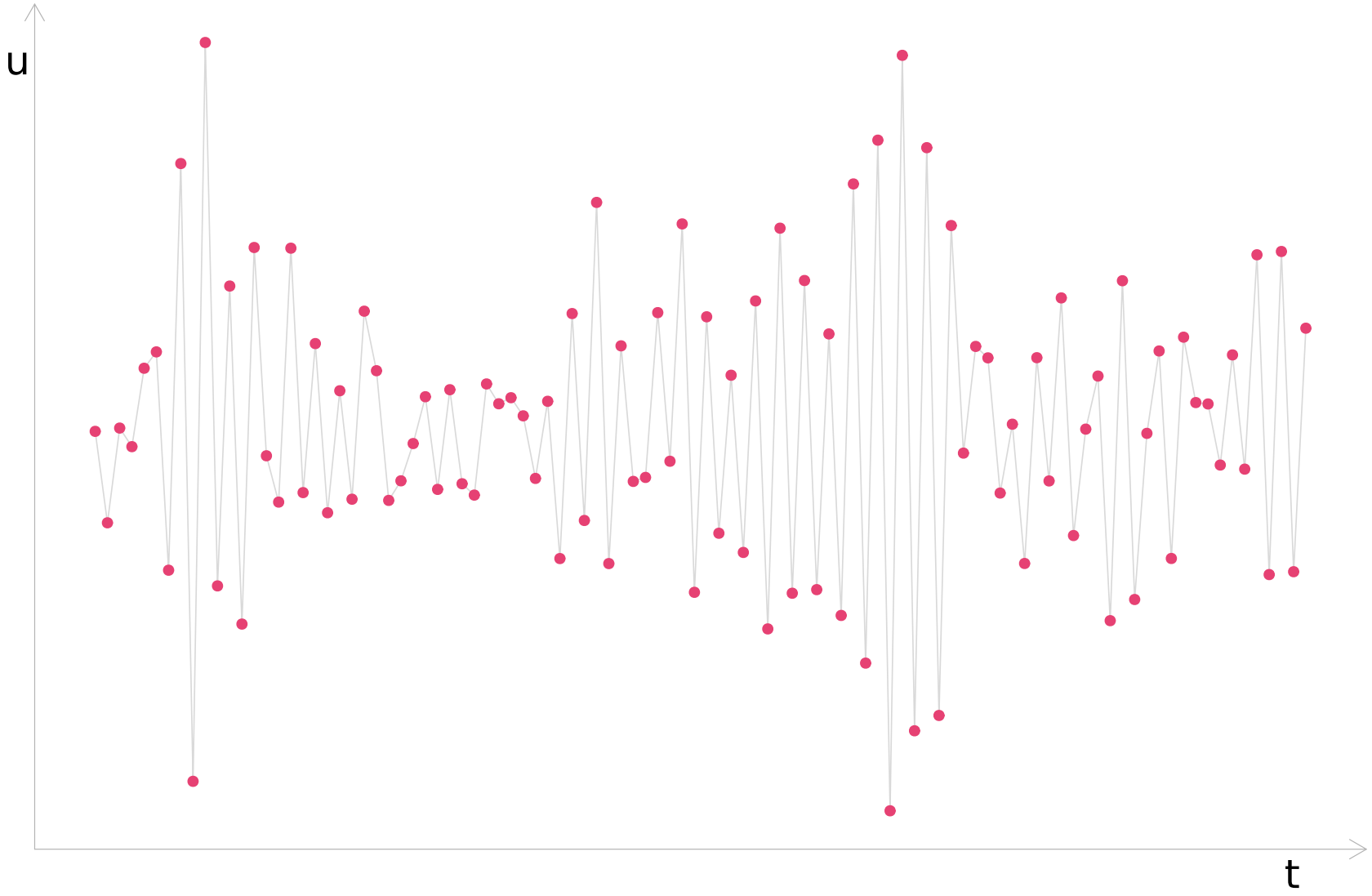
Positive autocorrelation: Disturbances (u_t) over time



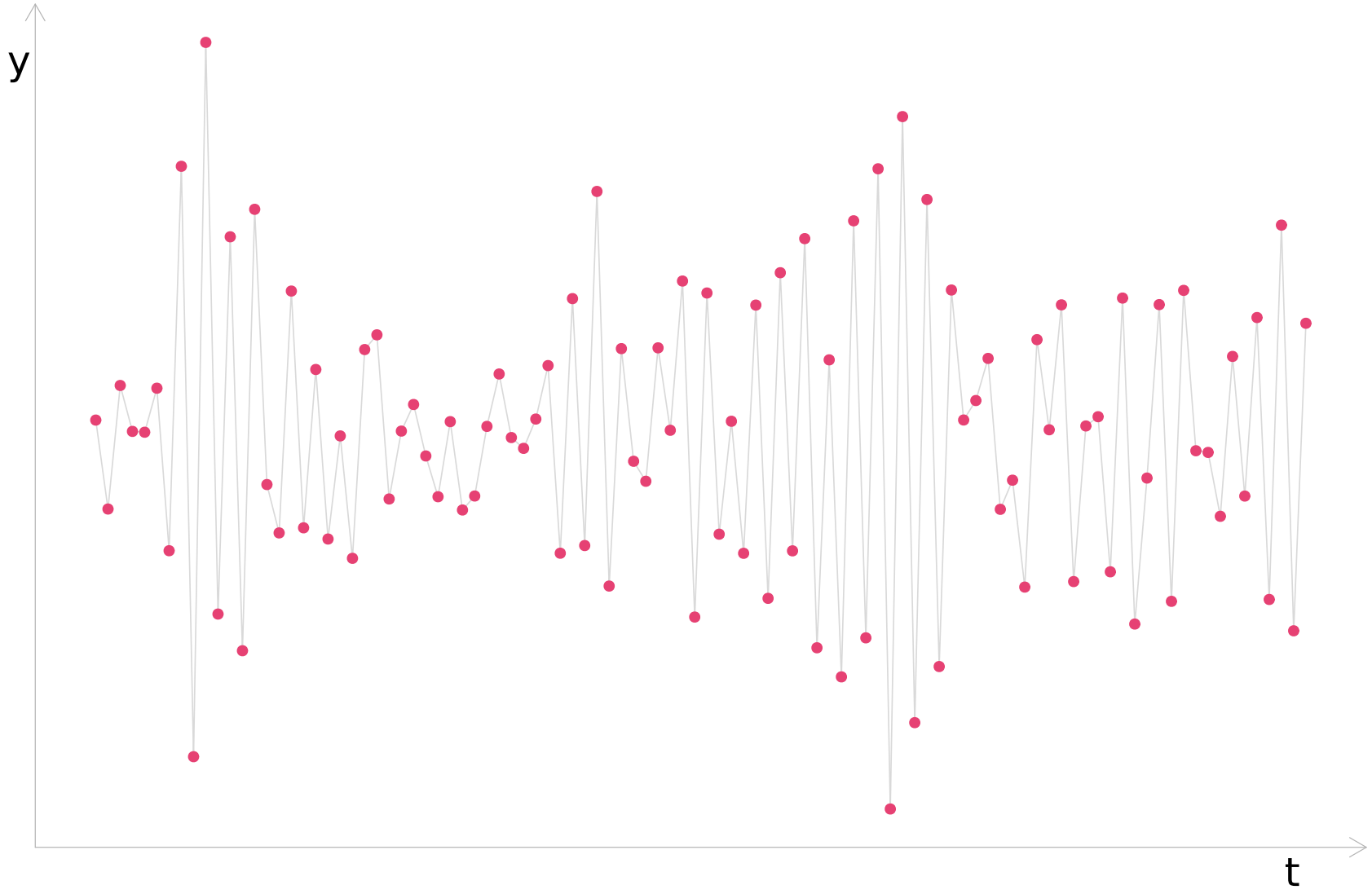
Positive autocorrelation: Outcomes (y_t) over time



Negative autocorrelation: Disturbances (u_t) over time



Negative autocorrelation: Outcomes (y_t) over time



Autocorrelation

In static time-series models

Let's start with a very common model: a static time-series model whose disturbances exhibit **first-order autocorrelation**, *a.k.a.* AR(1):

$$\text{Births}_t = \beta_0 + \beta_1 \text{Income}_t + u_t$$

where

$$u_t = \rho\, u_{t-1} + \varepsilon_t$$

and the ε_t are independently and identically distributed (*i.i.d.*).

Autocorrelation

In static time-series models

An AR(p) model/process has a disturbance structure of

$$u_t = \sum_{j=1}^{p} \rho_j u_{t-j} + \varepsilon_t$$

allowing the current disturbance to correlated with up to p of its lags.

Autocorrelation

OLS

For **static models** or **dynamic models with lagged explanatory variables**, in the presence of autocorrelation

1. OLS provides **unbiased** estimates for the coefficients.
2. OLS creates **biased** estimates for the standard errors.
3. OLS is **inefficient**.

Recall: Same implications as heteroskedasticity and correlated disturbances.

(Correlated disturbances are actually what we have with autocorrelation.)

Autocorrelation

OLS and lagged outcome variables

Consider a model with one lag of the outcome variable—ADL(1, 0)—model with AR(1) disturbances

$$\mathbf{Births}_t = \beta_0 + \beta_1 \mathbf{Income}_t + \beta_2 \mathbf{Births}_{t-1} + u_t$$

where

$$u_t = \rho u_{t-1} + \varepsilon_t$$

Autocorrelation

OLS and lagged outcome variables

To see this problem, first write out the model for t and $t-1$:

$$\begin{aligned}\text{Births}_t &= \beta_0 + \beta_1 \text{Income}_t + \beta_2 \text{Births}_{t-1} + u_t \\ \text{Births}_{t-1} &= \beta_0 + \beta_1 \text{Income}_{t-1} + \beta_2 \text{Births}_{t-2} + u_{t-1}\end{aligned}$$

and now note that $u_t = \rho u_{t-1} + \varepsilon_t$. Substituting...

$$\text{Births}_t = \beta_0 + \beta_1 \text{Income}_t + \beta_2 \text{Births}_{t-1} + \overbrace{(\rho u_{t-1} + \varepsilon_t)}^{u_t} \tag{1}$$

$$\text{Births}_{t-1} = \beta_0 + \beta_1 \text{Income}_{t-1} + \beta_2 \text{Births}_{t-2} + u_{t-1} \tag{2}$$

In (1), we can see that u_t depends upon (covaries with) u_{t-1}.
In (2), we can see that Births_{t-1}, a regressor in (1), also covaries with u_{t-1}.

Autocorrelation

OLS and lagged outcome variables

Implications: For models with **lagged outcome variables** and **autocorrelated disturbances**

1. The models **violate contemporaneous exogeneity**.
2. OLS is **biased and inconsistent** for the coefficients.

Autocorrelation

OLS and lagged outcome variables

Intuition? Why is OLS inconsistent and biased when we violate exogeneity?

Think back to omitted-variable bias...

$$y_t = \beta_0 + \beta_1 x_t + u_t$$

When $\mathbf{Cov}\left(x_t,\, u_t\right) \neq 0$, we cannot separate the effect of u_t on y_t from the effect of x_t on y_t. Thus, we get inconsistent estimates for β_1.

Autocorrelation and bias

Simulation

To see how this bias can look, let's run a simulation.

$$y_t = 1 + 2x_t + 0.5y_{t-1} + u_t$$
$$u_t = 0.9u_{t-1} + \varepsilon_t$$

One (easy) way generate 100 disturbances from AR(1), with $\rho = 0.9$:

```
arima.sim(model = list(ar = c(0.9)), n = 100)
```

Autocorrelation and bias

Simulation

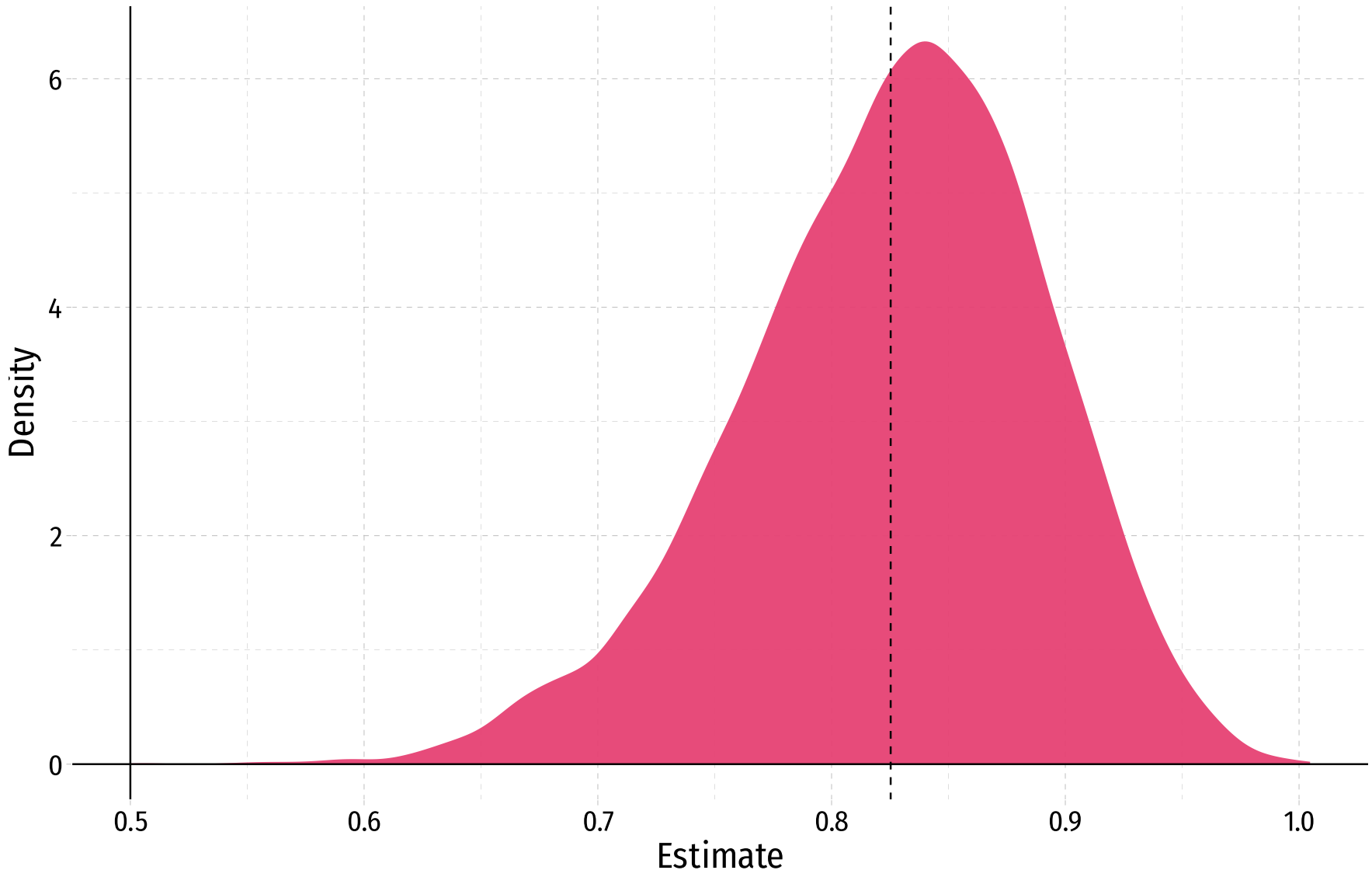
Outline of our simulation:

1. Generate T=100 values of x
2. Generate T=100 values of u
   - Generate T=100 values of ε
   - Use ε and ρ=0.9 to calculate $u_t = \rho\, u_{t-1} + \varepsilon_t$
3. Calculate $y_t = \beta_0 + \beta_1\, x_t + \beta_2\, y_{t-1} + u_t$
4. Regress y on x; record estimates

Repeat 1-4 10,000 times

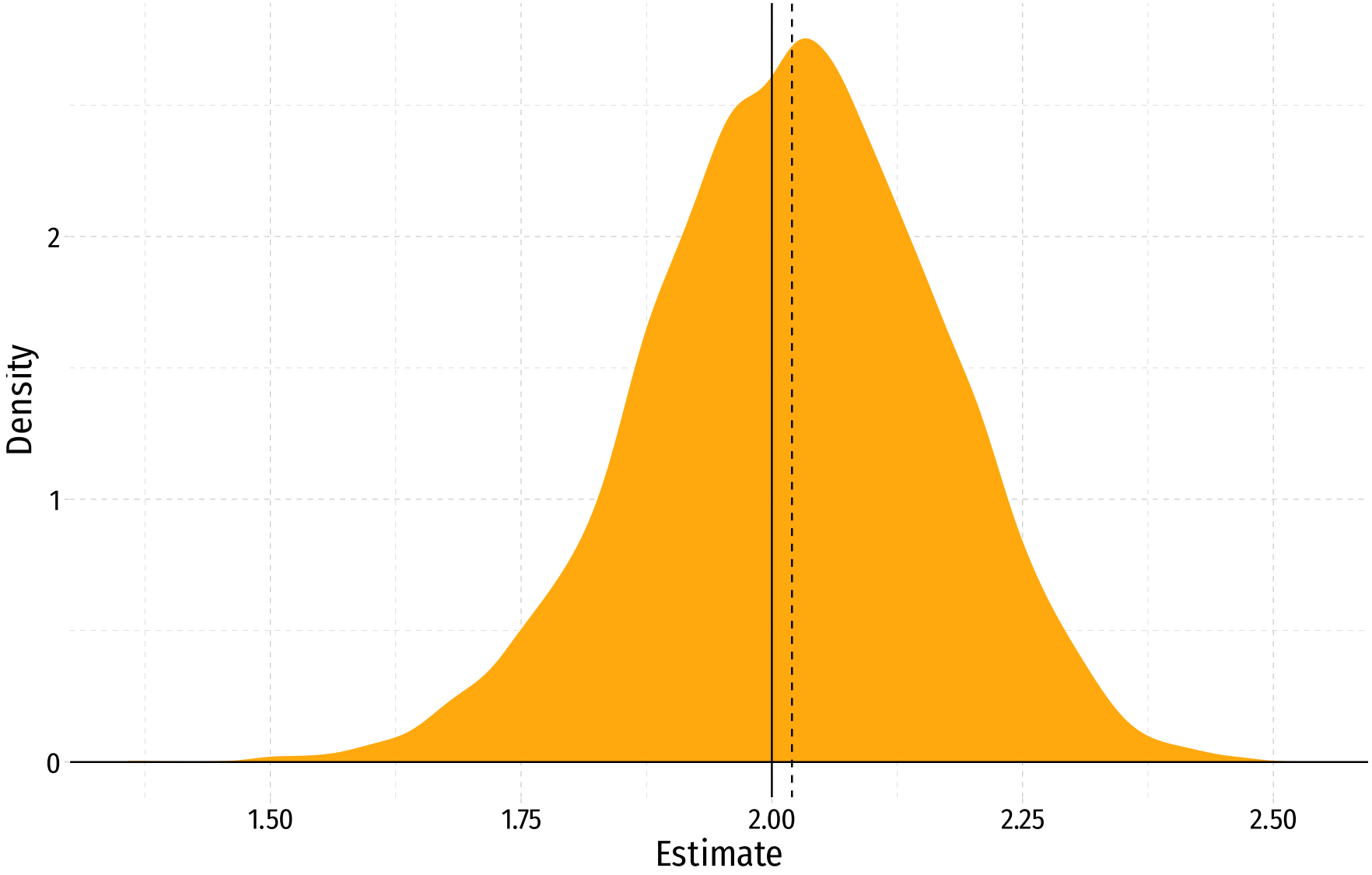
Distribution of OLS estimates, $\hat{\beta}_2$

$$y_t = 1 + 2x_t + 0.5y_{t-1} + u_t$$



Distribution of OLS estimates, $\hat{\beta}_1$

$$y_t = 1 + 2x_t + 0.5y_{t-1} + u_t$$



Testing for autocorrelation

Testing for autocorrelation

Static models

Suppose we have the **static model**,

$$\text{Births}_t = \beta_0 + \beta_1 \text{Income}_t + u_t \tag{A}$$

and we want to test for an AR(1) process in our disturbances u_t.

Testing for autocorrelation

Static models

Specifically, to test for AR(1) disturbances in the static model

$$\text{Births}_t = \beta_0 + \beta_1 \text{Income}_t + u_t \tag{A}$$

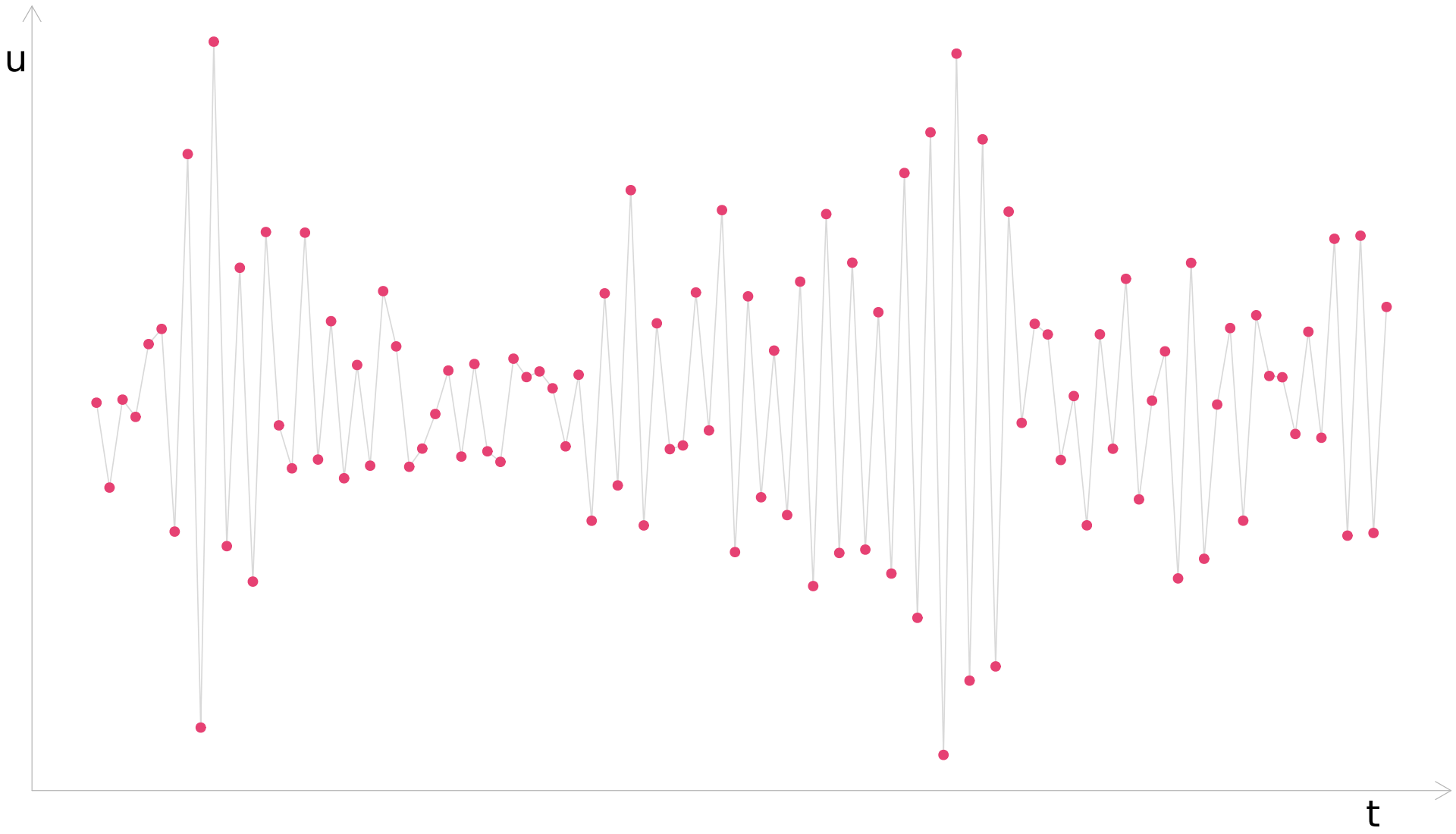
1. Estimate (A) via OLS.
2. Calculate residuals from the OLS regression in step 1.
3. Regress the residuals on their lags (without an intercept).

$$e_t = \rho e_{t-1} + v_t$$

4. Use a t test to determine whether there is statistically significant evidence that ρ differs from zero.
5. Rejecting H_0 implies significant evidence of autocorrelation.

For an example, let's return to our plot of negative autocorrelation.

Negative autocorrelation: Disturbances (u_t) over time



Testing for autocorrelation

Example: Static model and AR(1)

Step 1: Estimate the static model $(y_t = \beta_0 + \beta_1 x_t + u_t)$ with OLS

```
reg_est <- lm(y ~ x, data = ar_df)
```

Testing for autocorrelation

Example: Static model and AR(1)

Step 4: t test for the estimated $(\hat{\rho})$ coefficient in step 3.

```
tidy(reg_resid)
```

```
# A tibble: 1 × 5
  term   estimate std.error statistic  p.value
  <chr>     <dbl>     <dbl>     <dbl>    <dbl>
1 lag(e)   -0.851    0.0535     -15.9 6.88e-29
```

Testing for autocorrelation

Example: Static model and AR(3)

What if we wanted to test for AR(3)?

- We add more lags of residuals to the regression in *Step 3*.
- We **jointly** test the significance of the coefficients (*i.e.*, $\mathbf{LM}$ or $\boldsymbol{F}$).

Let's do it.

Testing for autocorrelation

Example: Static model and AR(3)

Step 1: Estimate the static model $(y_t = \beta_0 + \beta_1 x_t + u_t)$ with OLS

```
reg_est <- lm(y ~ x, data = ar_df)
```

Testing for autocorrelation

Example: Static model and AR(3)

Step 4: Calculate the $\mathbf{LM} = n \times R_e^2$ test statistic—distributed χ_k^2.

k is the number of regressors in the regression in *Step 3* (here, $k = 3$).

```
# Grab R squared
r2_e <- summary(reg_ar3)$r.squared
# Calculate the LM test statistic: n times r2_e
(lm_stat <- 100 * r2_e)
```

```
[1] 72.38204
```

```
# Calculate the p-value
(pchisq(q = lm_stat, df = 3, lower.tail = F))
```

```
[1] 1.318485e-15
```

Testing for autocorrelation

Example: Static model and AR(3)

Step 5: Conclude.

Testing for autocorrelation

Dynamic models with lagged outcome variables

Recall: OLS is biased and inconsistent when our model has *both*

1. a lagged dependent variable
2. autocorrelated disturbances

Testing for autocorrelation

Dynamic models with lagged outcome variables

Specifically, to test for AR(2) disturbances in the ADL(1, 0) model

$$\text{Births}_t = \beta_0 + \beta_1 \text{Income}_t + \beta_2 \text{Births}_{t-1} + u_t \tag{B}$$

1. Estimate (B) via OLS.
2. Calculate residuals (e_t) from the OLS regression in step 1.
3. Regress residuals on an intercept, explanatory variables, and lagged residuals.

$$e_t = \gamma_0 + \gamma_1 \text{Income}_t + \gamma_3 \text{Births}_{t-1} + \rho_1 e_{t-1} + \rho_2 e_{t-2} + v_t$$

4. Conduct LM or F test for $\rho_1 = \rho_2 = 0$.
5. Rejecting H_0 implies significant evidence of AR(2).

Testing for autocorrelation

Dynamic models with lagged outcome variables

For an example, let's consider the relationship between monthly presidential approval ratings and oil prices during President George W. Bush's[1] presidency.

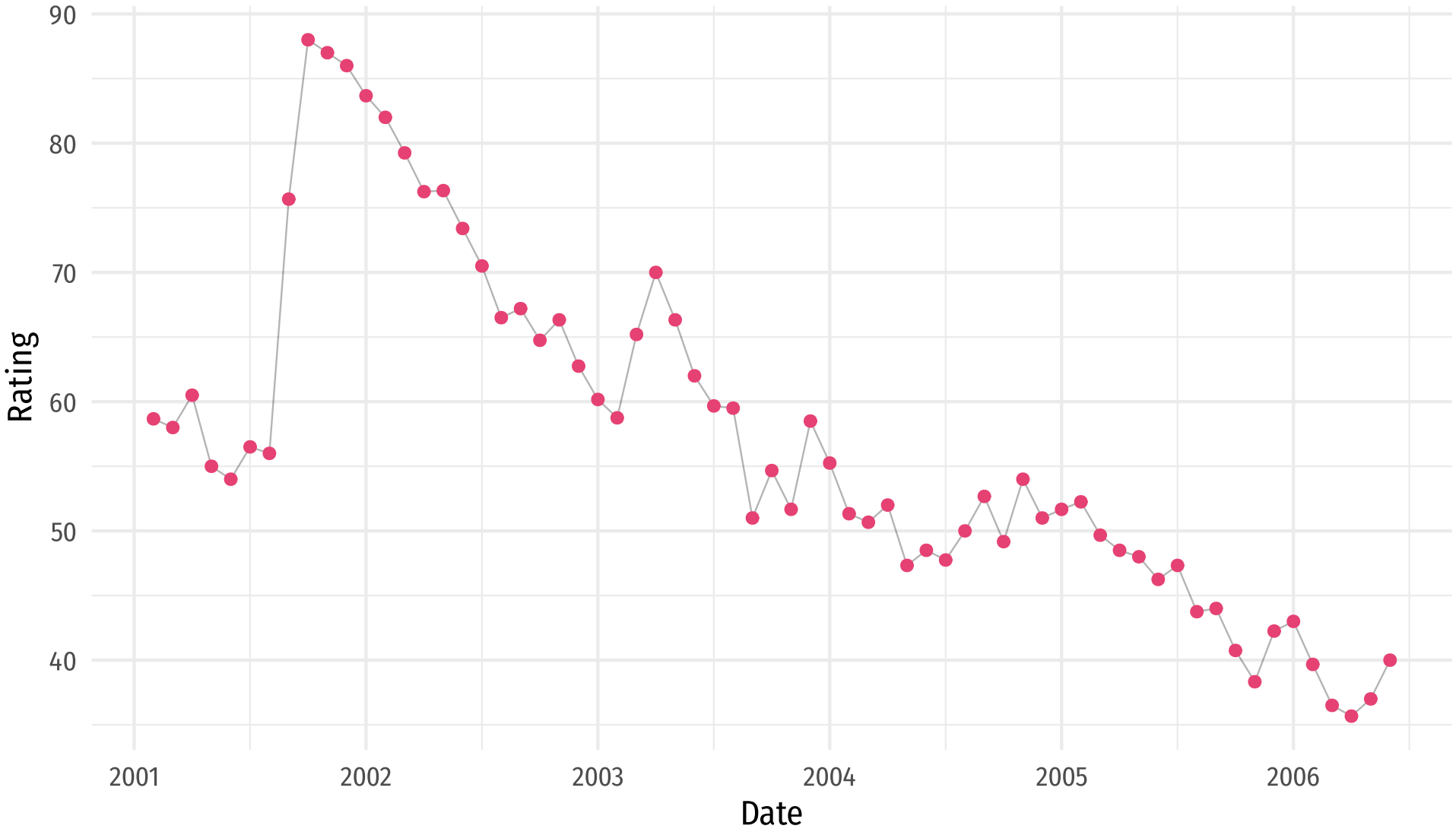
We will

1. specify the process as ADL(1, 0),
2. test for an AR(2) process in our disturbances.

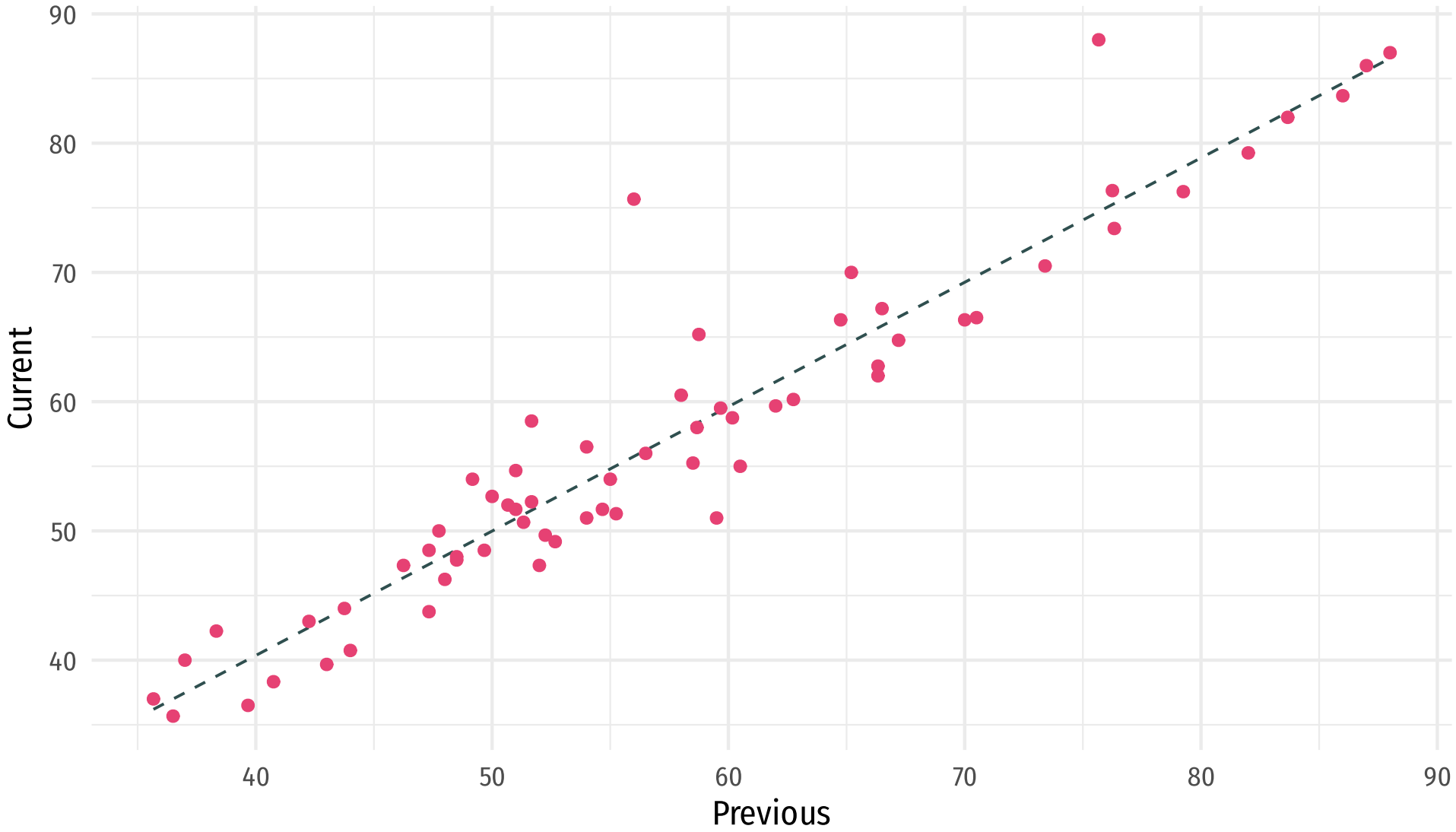
$$\text{Approval}_t = \beta_0 + \beta_1 \text{Approval}_{t-1} + \beta_2 \text{Price}_t + u_t$$

1. Fun with approval ratings.

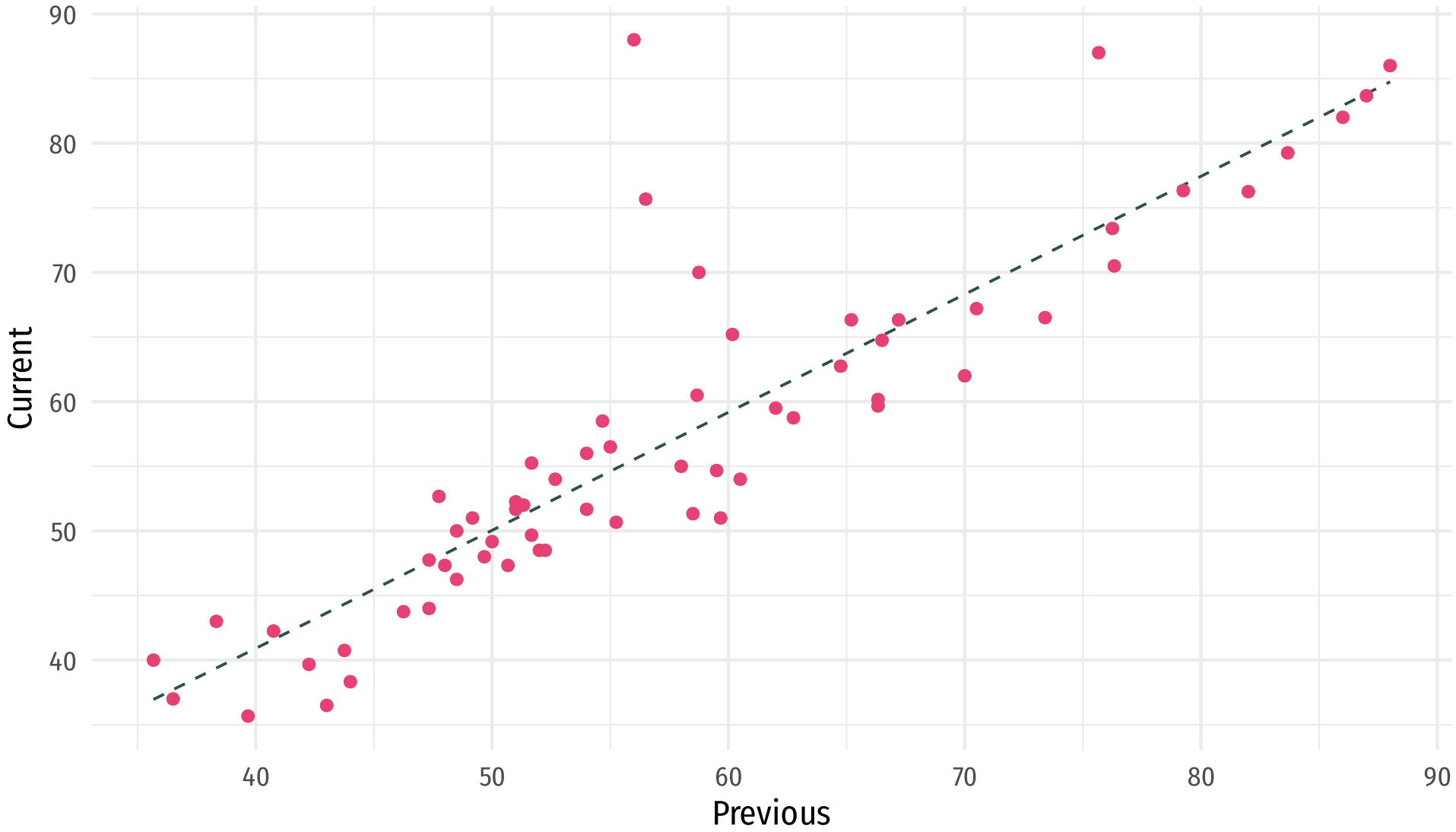
Monthly presidential approval ratings, 2001–2006



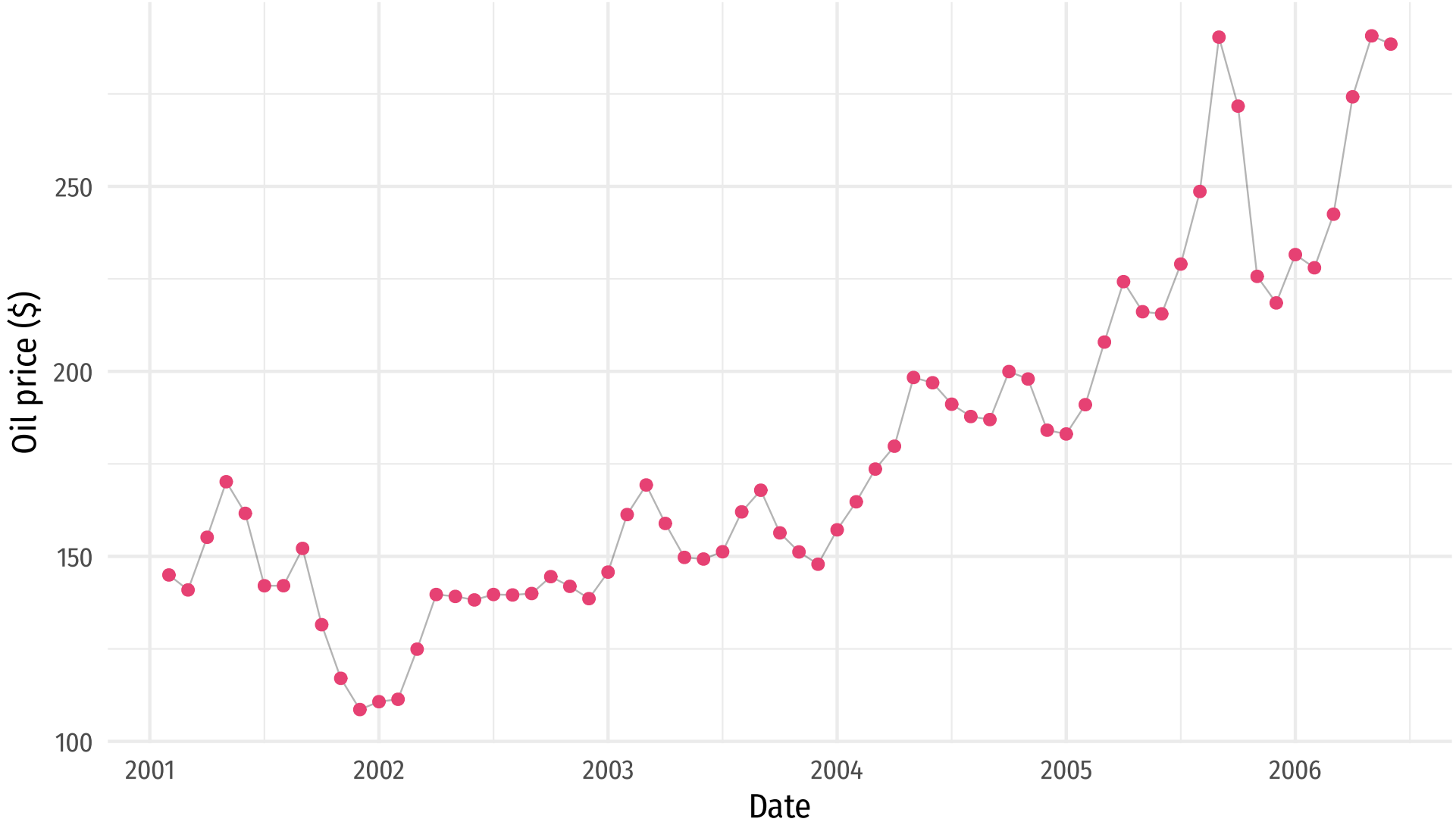
Approval rating vs. its one-month lag, 2001–2006



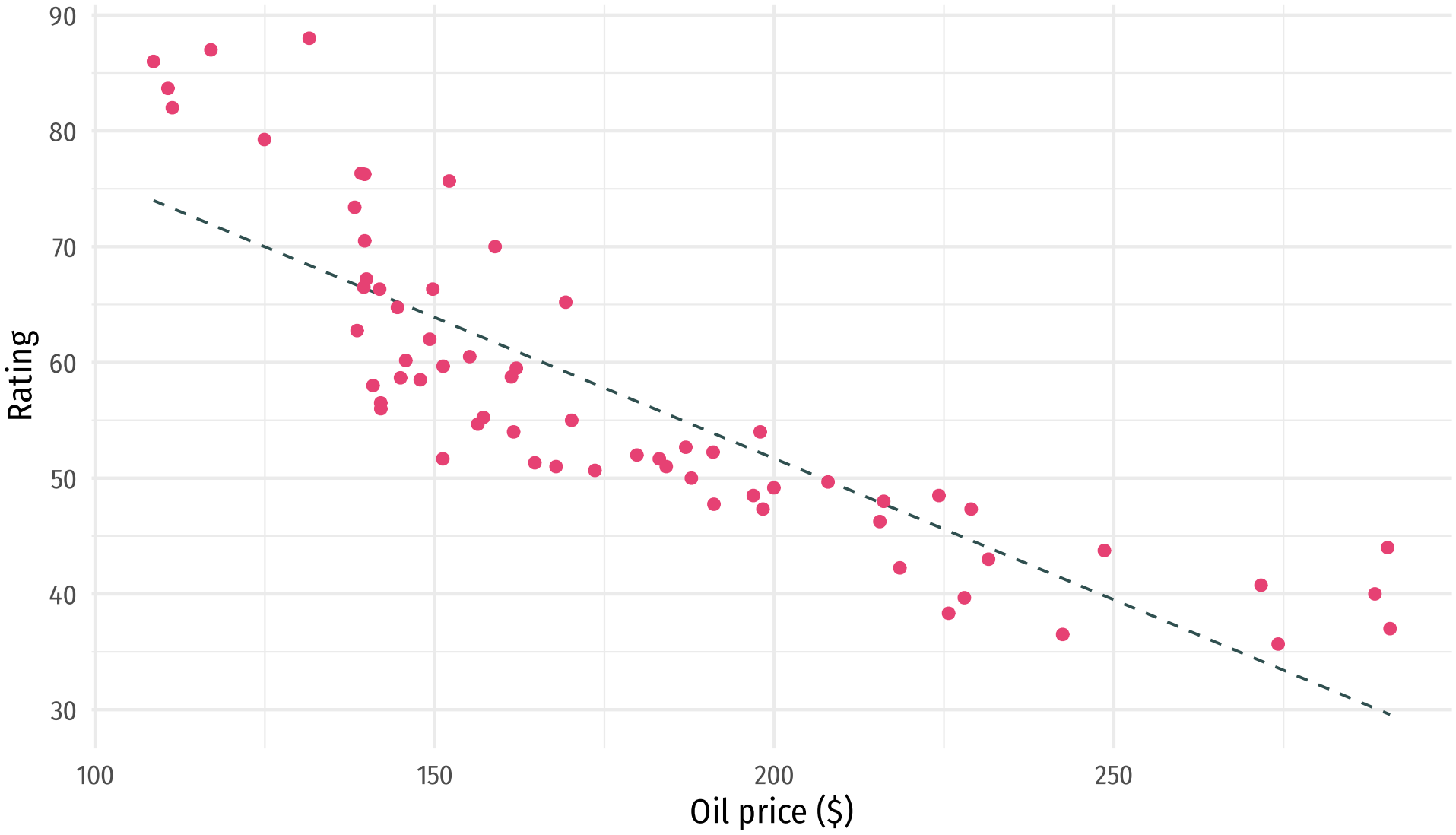
Approval rating vs. its two-month lag, 2001–2006



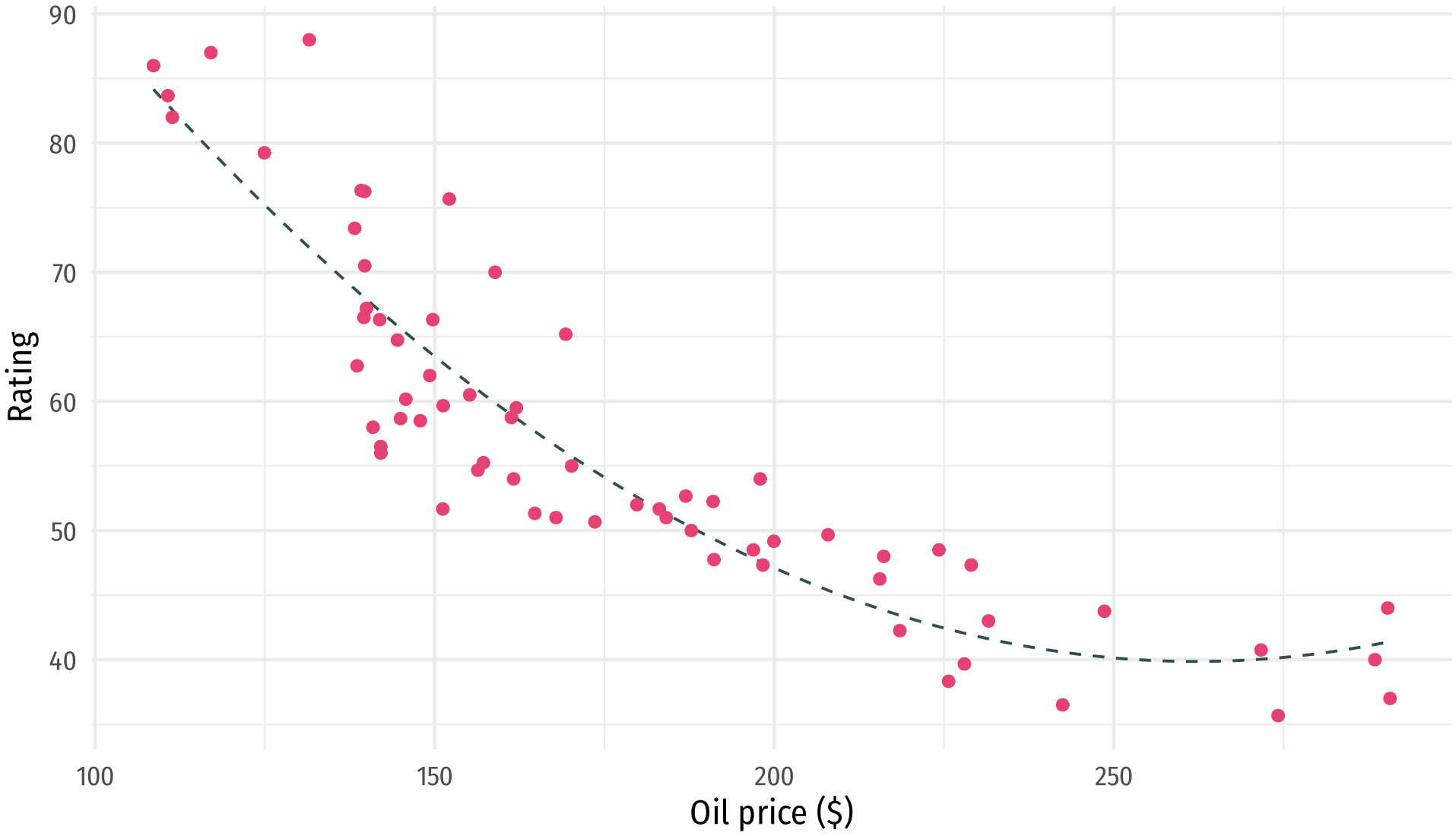
Oil prices, 2001–2006



Approval rating vs. oil prices, 2001–2006



Approval rating vs. oil prices, 2001–2006 with quadratic fit



Testing for autocorrelation

Example: Approval ratings and oil prices

Step 1: Estimate our ADL(1, 0) model with OLS.

```r
# Estimate the model
ols_est <- lm(
  approve ~ lag(approve) + price_oil,
  data = approval_df
)
# Summary
tidy(ols_est)
```

```
# A tibble: 3 × 5
  term         estimate std.error statistic  p.value
  <chr>           <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)   16.2       7.86        2.06 4.40e- 2
2 lag(approve)   0.841     0.0752     11.2  2.17e-16
3 price_oil     -0.0410    0.0215     -1.90 6.15e- 2
```

Testing for autocorrelation

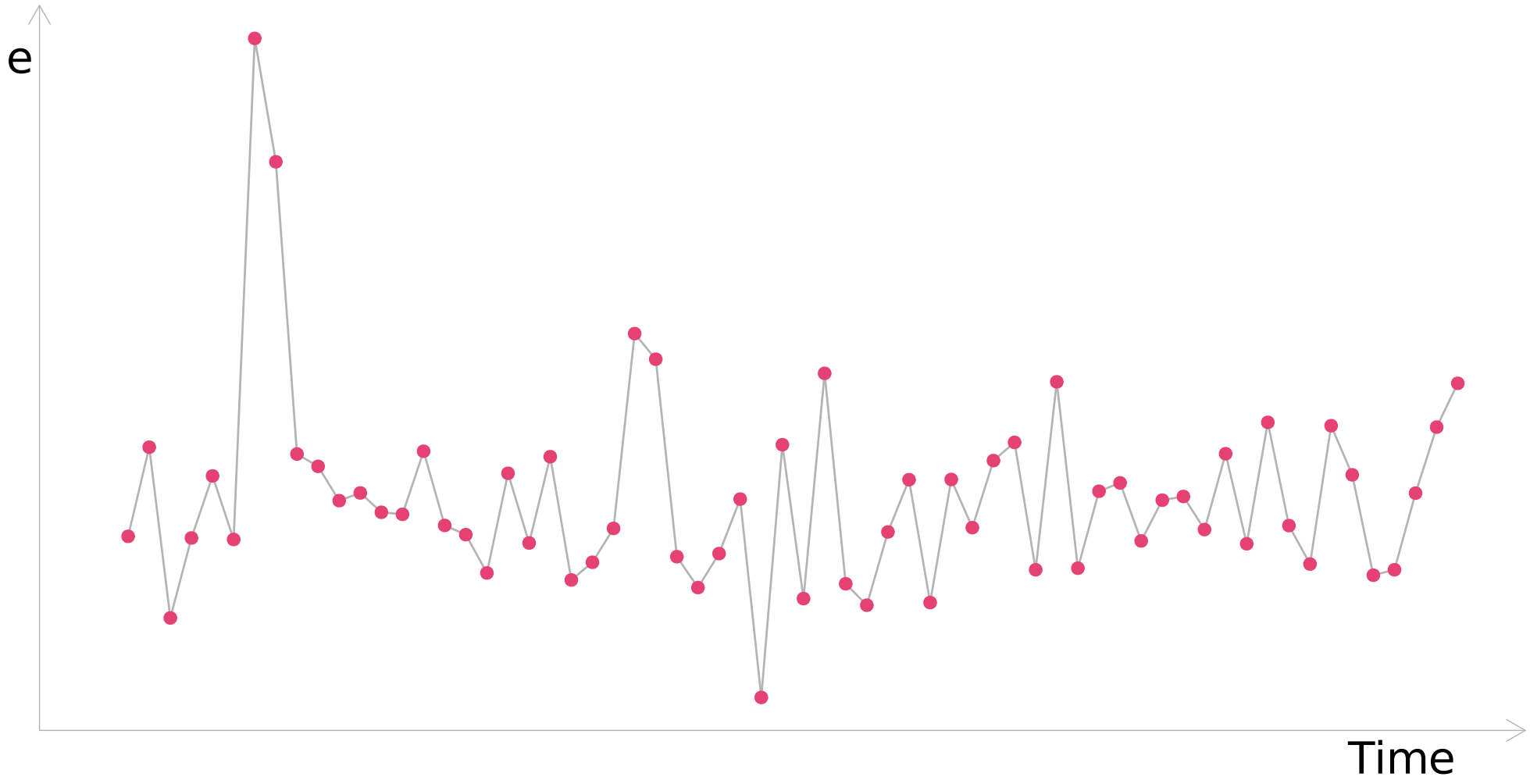
Example: Approval ratings and oil prices

Step 2: Record residuals from the OLS regression.

```
# Grab residuals
approval_df$e <- c(NA, residuals(ols_est))
```

Testing for autocorrelation

Example: Approval ratings and oil prices



Testing for autocorrelation

Example: Approval ratings and oil prices

Step 3: Regress residuals on an intercept, the explanatory variables, and lagged residuals.

```r
# BG regression
bg_reg <- lm(
  e ~ lag(approve) + price_oil + lag(e) + lag(e, 2),
  data = approval_df
)
```

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   7.92474    9.30455   0.852   0.3979
lag(approve) -0.08503    0.09192  -0.925   0.3589
price_oil    -0.01690    0.02407  -0.702   0.4854
lag(e)        0.25236    0.14648   1.723   0.0903 .
lag(e, 2)     0.07865    0.14471   0.544   0.5889
```

Testing for autocorrelation

Example: Approval ratings and oil prices

Step 4: F (or LM) test for $\rho_1 = \rho_2 = 0$.

Recall: We can test joint significance using an F test that compares the restricted (here: $\rho_1 = \rho_2 = 0$) and unrestricted models.

$$F_{q,\, n-p} = \frac{\left(\text{SSE}_r - \text{SSE}_u\right)/q}{\text{SSE}_u/\left(n-p\right)}$$

where q is the number of restrictions and p is the number of parameters in our unrestricted model (include the intercept).

We can use the `waldtest()` function from the `lmtest` package for this test.

Testing for autocorrelation

Example: Approval ratings and oil prices

Step 4: F (or LM) test for $\rho_1 = \rho_2 = 0$.

```
# BG regression
bg_reg <- lm(
  e ~ lag(approve) + price_oil + lag(e) + lag(e, 2),
  data = approval_df
)
# Test significance of the lags using 'waldtest' from 'lmtest' package
p_load(lmtest)
waldtest(bg_reg, c("lag(e)", "lag(e, 2)"))
```

Here, we're telling `waldtest` to test

- the model we specified in `bg_reg` (our **unrestricted model**)
- against a model without `lag(e)` and `lag(e, 2)` (our **restricted model**)

Testing for autocorrelation

Example: Approval ratings and oil prices

Step 4: F (or LM) test for $\rho_1 = \rho_2 = 0$.

```r
# BG regression
bg_reg <- lm(
  e ~ lag(approve) + price_oil + lag(e) + lag(e, 2),
  data = approval_df
)
# Test significance of the lags using 'waldtest' from 'lmtest' package
p_load(lmtest)
waldtest(bg_reg, c("lag(e)", "lag(e, 2)"))
```

```
Wald test

Model 1: e ~ lag(approve) + price_oil + lag(e) + lag(e, 2)
Model 2: e ~ lag(approve) + price_oil
  Res.Df Df      F Pr(>F)
1     57
2     59 -2 1.6153 0.2078
```

Testing for autocorrelation

Example: Approval ratings and oil prices

Step 5: Conclusion of hypothesis test

With a p-value of ~ 0.208, **we fail to reject the null hypothesis.**

- We cannot reject $\rho_1 = \rho_2 = 0$.
- We cannot reject "no autocorrelation".

Living with autocorrelation

Autocorrelation

Working with it

Suppose we believe autocorrelation is present. What do we do?

I'll give you three options.[1]

1. **Misspecification**
2. **Serial-correlation robust standard errors** (a.k.a. *Newey-West*)
3. **FGLS**

These solutions should look familiar.

1. You should take EC 422 to go much deeper into time-series analysis/forecasting.

Autocorrelation

Option 1: Misspecification

Misspecification with autocorrelation is very similar to our discussion with heteroskedasticity.

By incorrectly specifying your model, you can create autocorrelation.

Omitting variables that are correlated through time will cause your disturbances to be correlated through time.

Autocorrelation

Option 1: Misspecification

Example: Suppose births depend upon income and previous births

$$\text{Births}_t = \beta_0 + \beta_1 \text{Births}_{t-1} + \beta_2 \text{Income}_t + u_t$$

Autocorrelation

Option 1: Misspecification

"Proof"

$$
\begin{aligned}
v_t &= \beta_2 \text{Income}_t + u_t \\
v_{t-1} &= \beta_2 \text{Income}_{t-1} + u_{t-1}
\end{aligned}
$$

$\text{Cov}\left(v_t,\, v_{t-1}\right)$

Autocorrelation

Option 2: Newey-West standard errors

We can just focus on estimating consistent standard errors (fixing inference) in the presence of autocorrelation.

As we did with the heteroskedasticity-robust standard errors.

These standard errors are called **serial-correlation robust standard errors** (or *Newey-West standard errors*).[1]

1. We are not going to derive the estimator for these standard errors.

```r
# Estimate with NW SEs
fixest::feols(
  approve ~ l(approve) + price_oil,
  data = approval_df,
  panel.id = ~ country + time_months,
  se = 'NW'
)
```

```r
# Compare the two sets of standard errors from fixest
fixest::etable(est_iid, est_nw)
```

```
                           est_iid             est_nw
Dependent Var.:            approve            approve

Constant            16.16* (7.860)    16.16** (5.863)
l(approve)      0.8405*** (0.0752) 0.8405*** (0.0497)
price_oil        -0.0410. (0.0215)  -0.0410* (0.0176)
--------------- ------------------ ------------------
S.E. type                     IID   Newey-West (L=2)
Observations                   64                 64
R2                        0.90722            0.90722
Adj. R2                   0.90417            0.90417
---
```

The p-values on the price of oil coefficient: 0.062 vs. 0.023.

Autocorrelation

Option 3: FGLS

If we do not have a lagged outcome variable, then feasible generalized least squares (**FGLS**) can give us efficient and consistent standard errors.

Autocorrelation

Option 3: FGLS

$$\begin{aligned}\text{Births}_t - \rho\text{Births}_{t-1} =& \beta_0 (1-\rho) + \\ & \beta_1 \text{Income}_t - \rho\beta_1 \text{Income}_{t-1} + \\ & u_t - \rho u_{t-1}\end{aligned}$$

Autocorrelation

Option 3: FGLS

$$\begin{aligned}\text{Births}_t - \rho\text{Births}_{t-1} =& \beta_0 (1-\rho) + \\ & \beta_1 \text{Income}_t - \rho\beta_1 \text{Income}_{t-1} + \\ & u_t - \rho u_{t-1}\end{aligned}$$

which gives us a very specific dynamic model

$$\begin{aligned}\text{Births}_t =& \beta_0 (1-\rho) + \rho\text{Births}_{t-1} + \\ & \beta_1 \text{Income}_t - \rho\beta_1 \text{Income}_{t-1} + \\ & \underbrace{u_t - \rho u_{t-1}}_{=\varepsilon_t} \\ =& \beta_0 (1-\rho) + \rho\text{Births}_{t-1} + \\ & \beta_1 \text{Income}_t - \rho\beta_1 \text{Income}_{t-1} + \varepsilon_t\end{aligned}$$

that happens to be **free of autocorrelation**.

Autocorrelation

Option 3: FGLS

This **transformed model** is free of autocorrelation.

$$\begin{aligned}\text{Births}_t =& \beta_0 (1-\rho) + \rho \text{Births}_{t-1} + \\ & \beta_1 \text{Income}_t - \rho \beta_1 \text{Income}_{t-1} + \varepsilon_t\end{aligned}$$

Q: How do we actually estimate this model? (We don't know ρ.)

Autocorrelation

Summary: Main ideas

Autocorrelation means the disturbances correlated through time,

$$\text{Cov}\left(u_t, u_s\right) \neq 0 \quad \text{for } t \neq s.$$

- For **static time-series models**, OLS *coefficient* estimates are **unbiased**[1], but OLS *standard errors* are **biased** and OLS is **inefficient**.
- For **lagged outcome variables**, OLS coefficient estimates are **biased and inconsistent**.

1. Assuming exogeneity

Autocorrelation

Summary: Testing/living with autocorrelation

Tests for autocorrelation involve regressing residuals on their lags.

- In **static models**: regress residuals on their lags.
- In **dynamic models with lagged outcomes**: use **Breusch-Godfrey** regress residuals on the original regressors *plus* lagged residuals.

Potential solutions/responses (akin to *heteroskedasticity*):

- Consider whether the model is **misspecified**.
- Use **Newey-West** standard errors.
- For static models with AR disturbances, **FGLS** can transform the model to remove autocorrelation.

Admin

1. Schedule
2. R showcase
    - `ggplot2`
    - Writing functions
3. Review: Time series

Autocorrelation

1. Introduction
2. In static models
3. OLS and bias/consistency
    - Static models
    - Dynamic models with lagged y
4. Simulation: Bias
5. Testing for autocorrelation
    - Static models
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6. Working with autocorrelation
    - Misspecification
    - Newey-West standard errors
    - FGLS